SIOP Case Study 2: Validation Report

A rigorous analysis and audit of the Case Study 2 SIOP reporting workbook, including snapshot transformations, KPI calculations, and Power BI guidelines.

1. Executive Summary

This report provides an in-depth validation of the SIOP (Sales, Inventory, and Operations Planning) forecast snapshot transformation. The objective of the exercise is to evaluate three forecast snapshot periods—**Resultant (2023 P12)**, **Lag 0 (2023 P11)**, and **Lag 1 (2023 P10)**—against actual performance across 15 periods (2023 P10 to 2024 P12) and 4 countries (A, B, C, and D).

2. Key Statistics

Metric Dimension	Value / Aggregate
Total Rows in Fact Table	**112,282**
Distinct Items	**10,697**
Countries Covered	**4 (A, B, C, D)**
Periods Covered	**15 (2023 P10 to 2024 P12)**
Total Actual Revenue	**$46,039,033.18**
Total Resultant Revenue Forecast (2023 P12)	**$4,416,892.62**
Total Lag 0 Revenue Forecast (2023 P11)	**$5,877,361.08**
Total Lag 1 Revenue Forecast (2023 P10)	**$4,852,188.30**
Total Actual Non-Revenue / Capex Quantity	**4,912 / 0**

3. Country Performance KPI Metrics

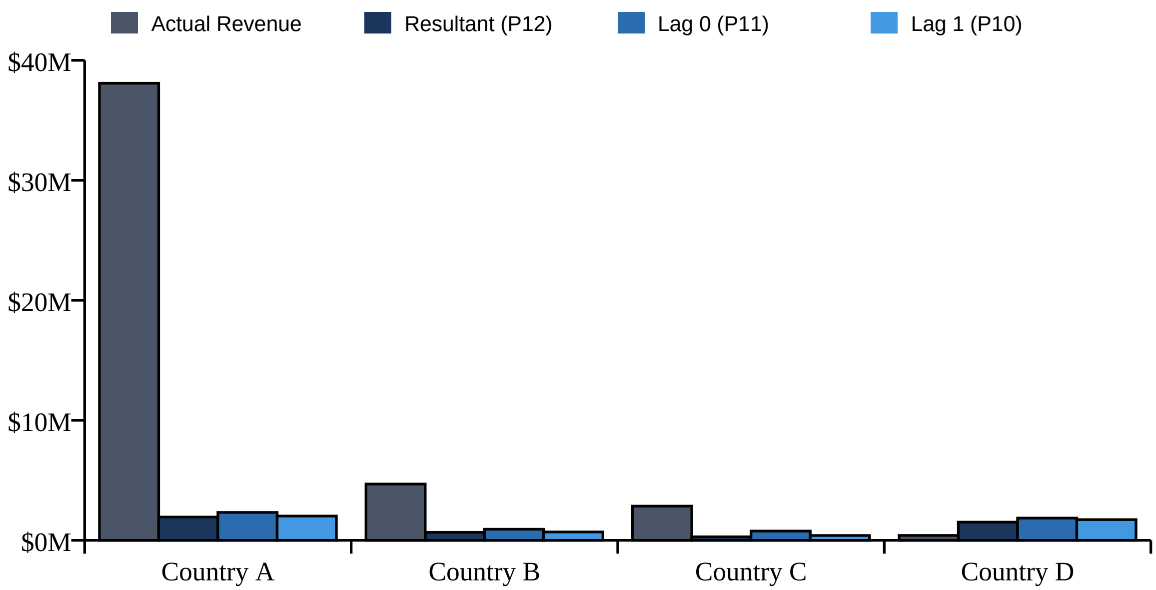
Using the standard forecast accuracy and bias formulas:
• **Accuracy %** = 1 - |Forecast - Actual| / Actual
• **Bias %** = (Forecast - Actual) / Actual
We aggregate the fact records by Country and calculate performance across all snapshots:

Country	Actual Rev	Resultant Acc / Bias	Lag 0 Acc / Bias	Lag 1 Acc / Bias

Country A	$38,088,468	5.1% / -94.9%	6.1% / -93.9%	5.3% / -94.7%
Country B	$4,690,326	14.2% / -85.8%	19.7% / -80.3%	14.9% / -85.1%
Country C	$2,851,024	10.3% / -89.7%	27.1% / -72.9%	14.2% / -85.8%
Country D	$409,216	-170.4% / 270.4%	-253.1% / 353.1%	-221.9% / 321.9%

4. Visualization: Revenue Comparison

Below is a side-by-side graphical comparison of Actual Revenue against the Resultant, Lag 0, and Lag 1 Forecast quantities across the countries:



5. Power BI Modeling Guide

For importing the dataset and reproducing the dashboard in Power BI, use the disconnected table method to allow the user to select the active forecast lag dynamically.

```
// Measure 1: Dynamic Revenue Forecast
Lag Revenue Forecast =
SWITCH(
    SELECTEDVALUE('Lag Selection'[Lag]),
    "Resultant", SUM(SIOP[Resultant_Revenue_Forecast]),
    "Lag 0", SUM(SIOP[Lag0_Revenue_Forecast]),
    "Lag 1", SUM(SIOP[Lag1_Revenue_Forecast])
)

// Measure 2: Revenue Forecast Accuracy %
Revenue Forecast Accuracy % =
1 - DIVIDE(
    ABS([Lag Revenue Forecast] - SUM(SIOP[Actual_Revenue])),
    SUM(SIOP[Actual_Revenue])
)

// Measure 3: Revenue Forecast Bias %
Revenue Forecast Bias % =
DIVIDE(
    [Lag Revenue Forecast] - SUM(SIOP[Actual_Revenue]),
    SUM(SIOP[Actual_Revenue])
```

```
)
```

6. Key Findings and Action Items

• **Under-Forecasting Trend:** Countries A, B, and C show massive under-forecasting bias (ranging from -72.9% to -94.9%). The current forecast snapshots are dramatically under-representing the actual volume of customer demand.
• **Over-Forecasting (Country D):** In contrast, Country D exhibits extreme positive bias (270.4% to 353.1%), where the forecast is roughly 3.7x to 4.5x actual values, leading to negative accuracy levels.
• **Systemic Bias Correction:** We recommend implementing a baseline correction using historical sales averages or seasonal factors to adjust raw snapshots before publishing them to executive dashboards.